National Highways Authority of India

INTERNSHIP AT NATIONAL HIGHWAY AUTHORITY OF INDIA
PUI GORAKHPUR

INTERNSHIP REPORT

PROJECT:4L Green Field Northern – East side Gorakhpur Bypass (26.616 Kms) starting from NH-29E (Chainage 83.308) and end at NH-28 (Chainage 285.100)

EPC Contractor: Vijai Construction (India) Pvt. Ltd.

Submitted by

Harsh Singh

(23BCE046)

Semester 6

Supervisors

Mr. Lalit Pratap Pal (Project Director)

Mr. Ruchir Agarwal (Manager Tech)

Mr. VK Pandey Sir (Team Leader)

DEPARTMENT OF CIVIL ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY HAMIRPUR

2025-26

DECLARATION

I , H A R S H S I N G H hereby declare that the internship entitled "**4L Green Field Northern – East side Gorakhpur Bypass (26.616 Kms) starting from NH-29E (Chainage 83.308) and end at NH-28 (Chainage 285.100)**" is an authentic record of work carried out by me during my training at "NHAI PIU GORAKHPUR "from 22/12/2025 to 16/01/2026 .

CERTIFICATE

This is to certify that Mr. Harsh Singh, Roll Number 23BCE046, a student of National Institute of Technology (NIT), Hamirpur, has successfully completed his four weeks of internship training from 22 December 2025 to 16 January 2026 at the National Highways Authority of India (NHAI).

During the training period, he was attached to the Project Implementation Unit (PIU), Gorakhpur, and worked on the NH-29E Greenfield 4-Lane Project (Gorakhpur Bypass). Throughout the internship, he was exposed to various aspects of highway engineering works including quality control procedures, pavement construction, structural works, surveying activities, and site execution practices.

His conduct, discipline, and performance during the training period were found to be sincere and satisfactory. We wish him success in his future academic and professional endeavours.

Date: 16/01/2026

Mr. Lalit Pratap Pal
(Project Director)

ACKNOWLEDGMENT

I would like to express my sincere gratitude to the National Highways Authority of India (NHAI) for providing me with the opportunity to undergo my internship training at the Project Implementation Unit (PIU), Gorakhpur, on the NH-29E Greenfield 4-Lane Project (Gorakhpur Bypass). The exposure provided by NHAI has been instrumental in helping me understand the practical aspects of highway engineering and large-scale infrastructure development.

I am deeply thankful to the officers, engineers, and staff of NHAI, PIU Gorakhpur, for their constant guidance, valuable technical inputs, and cooperative support throughout the training period. Their willingness to share professional knowledge and field experience greatly enhanced my understanding of construction practices, quality control procedures, and project execution under the EPC mode.

I would also like to acknowledge the support and assistance extended by the site engineers, supervisors, and laboratory personnel associated with the project. Their guidance during site visits, laboratory testing, surveying activities, and construction operations helped me gain practical insight into real-time field challenges and their solutions.

I express my sincere thanks to the faculty members of the Department of Civil Engineering, National Institute of Technology (NIT), Hamirpur, for their academic guidance, encouragement, and support throughout the internship period. Their continuous motivation played an important role in enabling me to successfully complete this training. Finally, I am grateful to my peers and all individuals who directly or indirectly contributed to making this internship a meaningful and enriching learning experience. The knowledge and skills gained during this period will serve as a strong foundation for my future academic and professional endeavours.

Harsh Singh
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ABSTRACT

The project involves the **construction of the 4L Green Field Northern – East side Gorakhpur Bypass** located in **Gorakhpur district, Uttar Pradesh**, under the **National Highways Authority of India (NHAI)** development program. The proposed bypass aims to decongest urban traffic within Gorakhpur city by providing an alternate route for through traffic, particularly heavy vehicles, while ensuring seamless regional connectivity.
The project encompasses a **design length of approximately 26.616 km**, starting from **NH-29E (Chainage 83.308)** and ending at **NH-28 (Chainage 285.100)**, designed to a **four-lane configuration with paved shoulders**, conforming to IRC and MoRTH standards for safety, geometry, and pavement performance.
The project is being implemented on an **Engineering, Procurement, and Construction (EPC)** basis. Upon completion, the **4L Green Field Northern – East side Gorakhpur Bypass** will substantially reduce congestion within the city, shorten travel times for through-traffic along the corridor, and stimulate **economic, logistical, and industrial growth** in the region by improving mobility, safety, and freight efficiency. It represents a critical infrastructure component for the long-term urban and transport planning of Gorakhpur and the adjoining areas.

INTRODUCTION TO NHAI

Overview of NHAI

The **National Highways Authority of India (NHAI)** is an autonomous organization under the **Ministry of Road Transport and Highways (MoRTH)**, Government of India. It is responsible for the development, maintenance, and management of national highways across the country. The authority was established with the objective of creating a safe, efficient, and world-class highway network to support India's economic growth and connectivity.

NHAI plays a crucial role in planning, designing, constructing, operating, and maintaining national highways, expressways, bridges, tunnels, and related infrastructure. The authority ensures that highway projects are executed in accordance with prescribed technical standards, quality norms, and safety regulations.

Objectives of NHAI

The primary objectives of the NHAI include:

Development and upgrading of national highways.

Maintenance of highway assets to ensure longevity and safety.

Implementation of projects using modern engineering practices and technology.

Improvement of road safety standards and user convenience.

Functions and Responsibilities

NHAI performs a wide range of critical functions to ensure the smooth operation of India's road network:

Planning and Execution: Strategic planning and execution of highway projects.

Project Monitoring: Monitoring of projects under various models including EPC, BOT, and HAM.

Quality Control: Ensuring adherence to quality standards as per MoRTH and IRC specifications.

Safety and Traffic Management: Implementing traffic management systems and ensuring safety compliance.

Note on Project Models:

EPC: Engineering, Procurement, and Construction

BOT: Build, Operate, and Transfer

HAM: Hybrid Annuity Model

Organizational Structure

NHAI operates through a well-defined hierarchical structure designed to facilitate efficient decision-making and project execution:

Headquarters: Led by the Chairman and Members.

Regional Offices: Managed by Regional Officers to oversee state-level operations.

Project Implementation Units (PIUs): Located at the project level for direct oversight.

Each **PIU** is headed by a Project Director, supported by a team of technical, contract, and quality engineers who ensure the smooth execution and monitoring of highway projects.

Importance of NHAI in Infrastructure Development

Highways form the backbone of a nation's transportation system. NHAI contributes significantly to national development by:

Enhancing regional and national connectivity.

Reducing travel time and transportation costs.

Supporting industrial growth and logistics efficiency.

Improving road safety standards across the network.

Strengthening economic integration across different states.

Through large-scale initiatives such as expressways, economic corridors, and access-controlled highways, NHAI has transformed India's road infrastructure to meet global standards.

Role of NHAI in Professional Training

NHAI provides valuable industrial exposure to engineering students through its internship programs. These internships are designed to bridge the gap between **theoretical knowledge** and **practical application**.

Participants gain real-time exposure to:

Highway construction practices.

Quality control procedures.

Contract management.

Site safety measures.

QUALITY TESTS

QUALITY TESTS

1. Soil Tests

1. Soil Sampling (Preparation of Dry Soil Sample)

While "Sampling" is a procedure rather than a "test" with a pass/fail limit, it is the first critical step for all other tests.
Apparatus: Riffle sampler (sample divider) or quartering tools (shovel/trowel), sieves (4.75mm, 19mm, 75mm), wooden mallet (for breaking clods), drying trays/oven.
Purpose: To obtain a **representative sample** from a large bulk of soil. Using the "Quartering" or "Riffling" method ensures the small sample tested in the lab accurately represents the tons of soil used on the road.
Acceptable Limit: N/A (Procedural Standard). However, the sample must be air-dried or oven-dried (max 60°C) before testing to prevent altering the soil properties.
IS Code: IS 2720 (Part 1)

2. Grain Size Analysis (Sieve Analysis)

Apparatus: Set of IS Sieves (4.75mm, 2mm, 425µ, 75µ, etc.), mechanical sieve shaker, balance, oven.
Purpose: To determine the particle size distribution (grading) of the soil, which helps in classifying the soil (Sand/Silt/Clay) and checking its suitability for drainage and filtration.
Acceptable Limit: Used for classification; generally, for subgrade, specific grading envelopes apply, but < 50% passing 75-micron sieve is preferred to avoid excess silt/clay.
IS Code: IS 2720 (Part 4)

3. Modified Proctor Compaction Test

Apparatus: Cylindrical metal mould (1000 cc), heavy metal rammer (4.9 kg with 450 mm drop), balance, oven, IS sieves (19mm, 4.75mm), mixing tools.
Purpose: To determine the **Optimum Moisture Content (OMC)** and **Maximum Dry Density (MDD)** of the soil. This data is used to set the compaction targets for the site rollers.
Acceptable Limit (MoRTH):
Subgrade: Minimum **97%** of MDD.
Embankment: Minimum **95%** of MDD.
IS Code: IS 2720 (Part 8)

QUALITY TESTS

4. California Bearing Ratio (CBR) Test

Apparatus: CBR mould (150mm dia), spacer disc, surcharge weights (2.5 kg each), loading machine (1.25 mm/min rate), penetration piston (50mm dia), dial gauges.
Purpose: To evaluate the **strength** and bearing capacity of the subgrade soil to determine the thickness of the road pavement layers.
Acceptable Limit (MoRTH):
Subgrade: Minimum **8%** (soaked value) for major highways (often varies by traffic, but 8-10% is standard for high-traffic NH).
General Fill: Minimum **2% - 5%** (often treated if lower).
IS Code: IS 2720 (Part 16)

5. Atterberg Limits (Liquid Limit & Plastic Limit)

Apparatus: Casagrande apparatus (or Cone Penetrometer), grooving tool, porcelain dish, spatula, balance, oven, glass plate.
Purpose: To classify the soil type (clay/silt/sand) and determine its consistency. It helps identify if the soil is too plastic (clayey) to be used.
Acceptable Limit (MoRTH):
Liquid Limit (LL): Maximum **50%** (for subgrade) to **70%** (for embankment).
Plasticity Index (PI): Generally should be less than **25%** for subgrade.
IS Code: IS 2720 (Part 5)

6. Free Swell Index (FSI) Test

Apparatus: 425-micron IS sieve, two 100ml graduated glass cylinders, balance, distilled water, kerosene.
Purpose: To determine the **expansive nature** of the soil (e.g., Black Cotton Soil). If soil swells too much when wet, it will damage the road.
Acceptable Limit (MoRTH):
Maximum **50%**. If FSI > 50%, the soil is considered "expansive" and typically rejected or treated before use in subgrade.
IS Code: IS 2720 (Part 40)

QUALITY TESTS

2. Aggregate Tests (Coarse)

1. Aggregate Impact Value (AIV) Test

•**Apparatus:** Impact testing machine (with 13.5-14 kg hammer), Cylindrical measure, Tamping rod, IS Sieves (12.5mm, 10mm, 2.36mm), Balance, Oven.
•**Purpose:** To determine the **toughness** of the aggregate and its resistance to sudden shock or impact (e.g., from moving traffic).
•**Acceptable Limit (MoRTH):**
- •**Sub-base (GSB/WMM):** Maximum **30%**.
- •**Base/Binder Course (DBM):** Maximum **27%**.
- •**Wearing Course (BC):** Maximum **24%**.

•**IS Code:** IS 2386 (Part 4)

2. Water Absorption Test

•**Apparatus:** Wire basket (perforated), Water tank, Dry soft absorbent cloths, Balance, Oven (100-110°C), Trays.
•**Purpose:** To measure the **porosity** of the stone. High water absorption indicates weak, porous rock that may break easily or absorb bitumen (leading to higher costs).
•**Acceptable Limit (MoRTH):**
- •Generally **Maximum 2%** for all road aggregates (especially for bituminous work).

•**IS Code:** IS 2386 (Part 3)

3. Specific Gravity & Apparent Specific Gravity

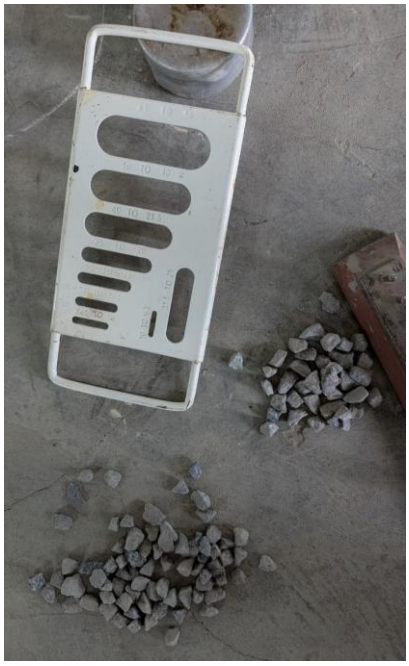
•**Apparatus:** Wire basket, Water tank, Balance (same setup as Water Absorption).
•**Purpose:**
- •**Specific Gravity:** Used in mix design calculations (voids analysis) to determine how much material is needed for a specific volume.
- •**Apparent Specific Gravity:** Considers only the impermeable volume of the rock.

•**Acceptable Limit:**
- •**No strict "Pass/Fail" limit**, but typical values for good road aggregate range between **2.5 to 2.9**.

•**IS Code:** IS 2386 (Part 3)

QUALITY TESTS

4. **Flakiness & Elongation Index (FI & EI)**



•**Apparatus:**
- •**Thickness Gauge** (for Flakiness).
- •**Length Gauge** (for Elongation).
- •IS Sieves (63mm to 6.3mm).

•**Purpose:** To check the **shape** of the particles.
- •*Flaky* (flat) and *Elongated* (long) particles are weak and break easily under rollers. We want "cubical" stones.

•**Acceptable Limit (MoRTH):**
- •**Combined (FI + EI):** Maximum **30%** (for DBM/BC/WMM).
- •*Note: Older specs had individual limits, but MoRTH 5th Revision focuses on the Combined Index.*

•**IS Code:** IS 2386 (Part 1)

3. Aggregate Tests (Fine)

Fineness Modulus (FM)

•**Apparatus:** Set of Standard IS Sieves (4.75mm, 2.36mm, 1.18mm, 600µm, 300µm, 150µm), Pan, Balance (0.1g accuracy), Oven (100-110°C), Mechanical Sieve Shaker.

•**Purpose:** To determine the **average particle size** of the sand.
- •It classifies sand as Fine, Medium, or Coarse.
- •It helps determine the **Grading Zone** (Zone I to IV). Zone II is typically preferred for concrete; Zone IV is too fine (like beach sand) and requires excessive cement.

•**Acceptable Limit (MoRTH / IS 383):**
- •**Fine Sand:** FM = **2.2 – 2.6**
- •**Medium Sand:** FM = **2.6 – 2.9** (Most common for highway concrete)
- •**Coarse Sand:** FM = **2.9 – 3.2**
- •*Note:* Sand with FM < 2.0 is generally rejected for structural works as it increases water demand significantly.

•**IS Code:**
- •**Test Method:** IS 2386 (Part 1)
- •**Specifications:** IS 383 (2016 Revision

QUALITY TESTS

4. Bitumen Tests

1. Penetration Test

Apparatus: Penetrometer (with standard needle), Water bath (maintained at 25.0 ± 0.1°C), Transfer dish, Timer, Thermometer.
Purpose: To measure the **hardness or consistency** of bitumen.
A weighted needle (100g) is allowed to sink into the bitumen for 5 seconds.
Higher value = Softer bitumen (e.g., used in cold climates).
Lower value = Harder bitumen (e.g., used in hot climates/heavy traffic).
Acceptable Limit (IS 73 / MoRTH):
VG-30: 50 to 70 (units in 1/10th mm).
VG-40: 40 to 60 (units in 1/10th mm).
IS Code: IS 1203

2. Softening Point Test (Ring and Ball Test)

Apparatus: Ring and Ball apparatus (Steel balls, brass rings), Beaker (glass container), Thermometer, Stirrer, Heater.
Purpose: To find the **temperature** at which bitumen changes from a semi-solid to a liquid state.
This ensures the road won't melt or bleed during hot Indian summers.
Acceptable Limit (IS 73 / MoRTH):
VG-30: Minimum **47°C**.
VG-40: Minimum **50°C**.
IS Code: IS 1205

3. Ductility Test

Apparatus: Ductility testing machine (briquette mould), Water bath (27°C).
Purpose: To measure the **elasticity** and adhesive properties of bitumen.
The sample is pulled apart at a standard speed (50 mm/min).
If ductility is too low, the road will crack under traffic loads (fatigue cracking).
Acceptable Limit (IS 73 / MoRTH):
VG-30: Minimum **40 cm**.
VG-40: Minimum **25 cm**.
IS Code: IS 1208

QUALITY TESTS

5. Concrete tests

1. Slump Cone Test (Workability)

Apparatus: Slump Cone (Frustum shape: Top 10cm, Bottom 20cm, Height 30cm), Tamping rod (16mm dia, 60cm long), Base plate, Scale.
Purpose: To measure the **workability** (fluidity) of fresh concrete.
Too stiff = Difficult to place/honeycombing.
Too wet = Low strength/segregation.
Acceptable Limit (MoRTH / IS 456):
Pavement (PQC): 25 ± 10 mm (Very stiff, laid by paver).
RCC / Bridges: 50 – 100 mm (Medium workability).
IS Code: IS 1199

2. Compressive Strength of Concrete (Cube Test)

Apparatus: Standard Cube Moulds (150 mm x 150 mm x 150 mm), Tamping rod, Curing tank, CTM (Compression Testing Machine).
Purpose: To determine the characteristic strength of the concrete (e.g., M40 means it bears 40 N/mm²). This is the **most important** test for payment and quality acceptance.
Acceptable Limit (IS 456):
7 Days: ~65-70% of Target Strength (Early check).
28 Days: Must achieve **100%** of Characteristic Strength (e.g., ≥ 40 MPa for M40).
IS Code: IS 516

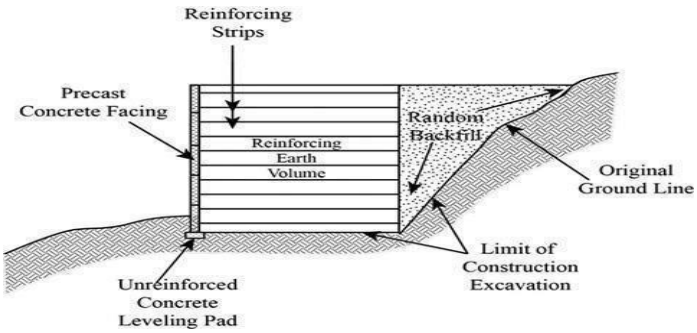
STRUCTURE

REINFORCED EARTH (RE)WALL SYSTEM

A reinforced earth wall is designed and constructed to resist the lateral pressure of the soil and supports the soil laterally so that it can be maintained at different levels on both sides. The lateral pressure could be also due to earth filling, liquid pressure, sand, and granular materials. The walls are used to bound soils between two different elevations often in areas of terrain possessing undesirable slopes. These walls are an economical way to meet everyday earth retention needs for highway and bridge grade separations, railroads and mass transit systems. They are also used in response to difficult design conditions such as very high structures, restricted space, where obstructions within the soil mass are present.

REINFORCED EARTH WALL

Construction of the Reinforced Earth structure consisting of the erection and positioning of facing panels, installation of joint materials, connection of reinforcing strips, placement and compaction of Select Granular Backfill.



COMPONENTS OF REINFORCED EARTH WALL

1. Components of RE Wall

- **Facia Panel/RE Panel: The minimum grade of concrete for the precast facia panel should be M35, with a minimum thickness of 180mm. Ethylene Propylene Diene Monomer (EPDM) Pads are used to maintain spacing between RE panels.**
- **Reinforced Fill: The fill material is borrowed from quarries, river beds, etc., and should be free of organic material. Key properties include grain size (finer than 75 microns less than 15%), plasticity index (PI < 6), and bulk density as per design requirements.**
- **Drainage Media/Filter Media: A drainage media of a minimum 600 mm width at the back of the RE panel facing is usually used to avoid the development of hydrostatic pressures.**
- **Reinforcement Material: Different types of reinforcements are used, either metallic elements like bars, strips, plates, or polymeric elements like strips, grids, rods, mesh. These reinforcements provide resistance to tensile strain.**

2. Geocomposite fibre behind RE wall

- **Geocomposite is a combined geosynthetic (typically geotextile + geonet or geogrid) placed behind the RE wall panels, mainly to provide drainage and filtration while also assisting in reinforcement.**

- **In RE walls it is usually laid vertically or inclined against the back of the facing panels, so that seepage water from the backfill is collected and safely drained down to the toe drain, reducing pore water pressure and preventing build-up of**

CONSTRUCTION METHODOLOGY FOR REINFORCED EARTH (RE) WALLS

Reinforced Earth (RE) walls, also known as Mechanically Stabilized Earth (MSE) walls, enhance soil strength through strategically placed reinforcements. Cost-effective and versatile for highway projects like Gorakhpur Bypass Road (Ring Road).

Site Preparation

Clear vegetation/debris. Grade to design levels. Remove unsuitable materials. Engineer approves foundation surface.

Foundation Construction

Excavate 6-12". Level/compact base. Add gravel layer. Install geotextile for drainage.

Reinforcement Installation

Place geogrid/metallic strips horizontally. Extend beyond Rankine zone. Maintain alignment/tensioning. Use levelling rods at base for panel alignment.

Backfill Placement

Use well-graded granular soil. Layered placement with compaction (vibrating rollers). Free of organics/rocks

Facing Construction

Concrete panels/modular blocks connected to reinforcement. Cover joints with geocomposite fibre strips for drainage.

Drainage Systems

Perforated pipes, geocomposites, weep holes prevent hydrostatic pressure buildup.

Quality Control

Test backfill/compaction. Inspect reinforcement placement. Monitor settlement/alignment during construction

Major Bridge Structure (Ch.6+550)

Introduction As part of the 4-Lane Greenfield Gorakhpur Bypass project (NH-29E to NH-28), the alignment necessitates the construction of Major Bridges to navigate significant hydrological obstacles (rivers/canals). Per the project contract and IRC:5 definitions, structures with a total linear length exceeding **60 meters** are classified as Major Bridges. These structures are critical components of the project, designed to ensure all-weather connectivity and hydraulic sufficiency.

Design & Structural Configuration The Major Bridges in this project are designed for a service life of **100 years**, complying with MoRTH specifications and relevant IRC codes (IRC:6, IRC:21, IRC:78, and IRC:112).
Foundation Type: Deep foundations are utilized, specifically **Bored Cast-in-Situ Reinforced Concrete Piles**, to ensure stability against scour and heavy axle loads typical of National Highways.
Substructure: The substructure consists of circular/rectangular piers and abutments supported on pile caps.
Superstructure: The deck system typically comprises **Pre-Stressed Concrete (PSC) I-Girders** or Box Girders, supporting an RCC deck slab. This design is chosen for its superior load-bearing capacity and durability.
Bearings & Joints:
Bearings: POT-PTFE or Elastomeric bearings are installed to accommodate thermal expansion and seismic movements.
Expansion Joints: Strip Seal or Modular expansion joints are used to ensure a smooth riding quality.

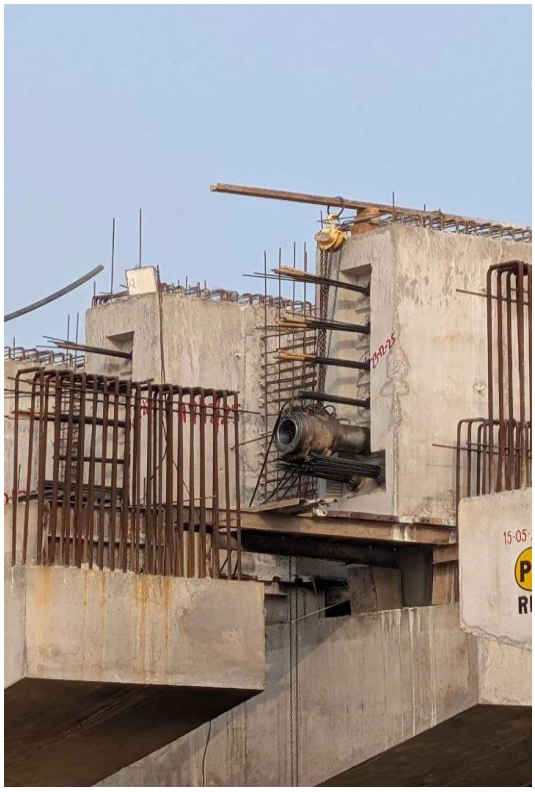
Materials Used
Concrete: High-performance concrete grades are strictly adhered to:
Piles: M-35 Grade
Pile Cap/Substructure: M-35 / M-40 Grade
Superstructure (PSC Girders): M-50 Grade
Steel: Fe-500D TMT bars are used for reinforcement, with High-Tensile Steel (HTS) strands used for prestressing the girders.

Pier, Pier Cap and Pile Cap

PTFE Bearings

Construction Methodology The construction sequence observed during the project execution follows this standard protocol:
Survey & Layout: Exact fixing of pile locations using Total Station.
Piling Work: Boring, lowering of reinforcement cage, and tremie concreting.
Pile Cap: Excavation, PCC, reinforcement binding, and casting of the cap.
Pier & Pier Cap: Shuttering and casting of the vertical pier shaft followed by the pier cap.
Girder Launching: Pre-casting of girders in the casting yard, stressing, and launching them onto the piers using cranes or launchers.
Deck Slab & Finishing: Casting the deck slab, installing crash barriers, and laying the bituminous wearing course.

Quality Control & Defects Liability Strict quality control measures are implemented during concreting (Slump test, Cube strength). As per the EPC agreement, the Major Bridge structures carry a **Defects Liability Period (DLP) of 5 years**, requiring the contractor to ensure structural integrity and perform routine maintenance.



Duct for Post Tensioning

Post Tensioning the Girder

Geostrap Belt, Panel Cavities, and Backfill Interaction

Geostrap belts were provided behind the RE panels as reinforcement elements. These straps interact with the backfill material to resist lateral earth pressure through frictional resistance.

- Panel cavities: Provided to accommodate geostrap connections and ensure proper anchorage.
- Slope of RE wall: A 1-degree inward slope was maintained to enhance stability and aesthetics.
- Geotextile at joints: Provided between panel joints to prevent loss of fine material through panel gaps.

A composite drainage layer was also provided behind the panels. This composite allows only water to drain out, while retaining soil particles, thereby preventing buildup of pore water pressure behind the wall.

Review of Structural Drawing of PSC Girder

During the site visit, the structural drawing of a PSC girder was studied. The drawing provided details of:

- Girder cross-section and dimensions
- Prestressing tendon layout
- Reinforcement detailing
- Bearing locations
- Deck slab interaction with girder

Understanding structural drawings helped in correlating design intent with actual site execution and construction sequencing.

Textile used at Joints

Geostrap Belts

HIGHWAY

HIGHWAY

A highway is a public road designed for the efficient and safe movement of vehicles and goods between cities, towns, and rural areas. It forms a crucial part of a nation's transportation infrastructure, supporting economic development, connectivity, and mobility. Highways are typically constructed with superior materials, engineered alignments, and controlled access features to handle high-speed, high- volume traffic. They include various components such as the carriageway, shoulders, medians, side drains, and embankments, all designed according to standards set by authorities like the Indian Roads Congress (IRC) or international agencies. Highways play a vital role in reducing travel time, enhancing logistics, and contributing to the overall growth of trade, tourism, and social integration. National Highways Authority of India (NHAI) and the National Highways and Infrastructure Development Corporation Limited (NHIDCL) are the authorities responsible for the development, maintenance and management of national highways in India.

1. Carriageway

The carriageway is the portion of the road surface that is intended for the movement of vehicles.

It comprises one or more lanes, depending on the traffic volume and classification of the road. Standard width in 4 lane width is typically 7 meters.

2. Shoulders

Shoulders are the strips provided on either side of the carriageway. They are used for emergency parking, vehicle breakdowns, and to enhance the lateral stability of the pavement. Shoulders may be paved or unpaved, depending on the road type and design standards.

3. Roadway Width

The roadway width includes the total width of the carriageway plus the shoulders on both sides. It represents he full width available for vehicular and marginal activities, excluding pedestrian pathways.

4. Camber (Cross Slope)

Camber is the transverse slope provided to the road surface, typically ranging between 2% to 3%, to facilitate the drainage of surface water. It prevents water stagnation on the pavement and enhances its longevity.

5. Kerbs

Kerbs are raised structural elements provided at the edges of the carriageway, especially in urban roads. They serve to confine the movement of vehicles within the designated carriageway and protect pedestrians.

6. Footpath (Sidewalk)

A footpath is a dedicated pedestrian walkway, usually located adjacent to the kerb in urban areas. It ensures the safety of pedestrians by separating them from vehicular traffic.

7. Median (Divider)

A median is a physical or painted separation between two directions of traffic flow. It enhances road safety by reducing head-on collisions and can also accommodate road furniture, signage, or plantation.

8. Right of Way (ROW)

The Right of Way is the total land width acquired for road construction and future expansion. It encompasses the carriageway, shoulders, footpaths, medians, drains, and any reserved area for utilities.

9. Side Slopes

Side slopes are provided on embankments or cuttings to ensure structural stability and to prevent soil erosion. Their steepness is defined by the type of soil and the height of the embankment or depth of the cutting.

10. Drainage System

Roadside drains, either open or closed, are designed to collect and dispose of stormwater. A proper drainage system is essential to protect the pavement structure from water-related damage and to maintain traffic safety.

Pavement Cross-Section

Pavement Type: The project utilizes **both Flexible and Rigid Pavements** depending on the location (main carriageway and toll plazas/structures).

1. Flexible Pavement Cross-Section (Main Carriageway)

Flexible pavement is typically used for the majority of the main highway length. The structure consists of multiple layers designed to distribute traffic loads to the subgrade.
Total Thickness: Typically **600 mm – 850 mm** (excluding subgrade).
Design Life: 20 Years.

Layer Composition (Top to Bottom):

Layer Name	Material	Approx. Thickness	Function
Wearing Course	**BC** (Bituminous Concrete)	**40 mm - 50 mm**	Provides smooth riding surface, skid resistance, and waterproofing.
Binder Course	**DBM** (Dense Bituminous Macadam)	**90 mm - 130 mm**	Main load-bearing bituminous layer; transfers load to the base.

Layer Name	Material	Approx. Thickness	Function
Base Course	**WMM** (Wet Mix Macadam)	**250 mm**	Crushed aggregate layer mixed with water; provides structural support.
Sub-Base	**GSB** (Granular Sub-Base)	**200 mm - 300 mm**	Drainage layer and transition zone to protect subgrade from fines.
Subgrade	**Compacted Soil**	**500 mm**	Foundation layer (CBR ≥ 8%).

Cross Section Features:
Lane Width: 2 x 7.0m (4-lane divided carriageway).
Paved Shoulder: 1.5m - 2.5m (often same composition as main carriageway up to DBM/Base).
Earthen Shoulder: 1.0m - 1.5m (Compacted earth).
Camber: Typically **2.5%** for drainage.

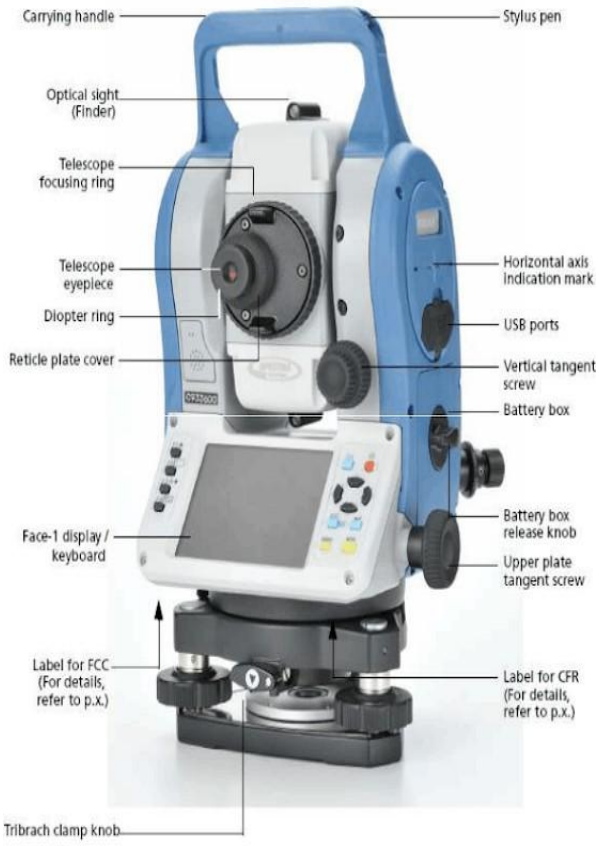
2. Rigid Pavement Cross-Section (Toll Plaza / Structures)

Rigid pavement (Concrete) is mandated for **Toll Plazas** (approx. 1 km length) and sometimes at **Wayside Amenities** or high-stress intersections to resist rutting and oil spillage damages.
Total Thickness: Typically **500 mm - 600 mm** (Concrete + DLC + GSB).
Design Life: 30 Years (Contract requires 10-year defect liability for rigid).

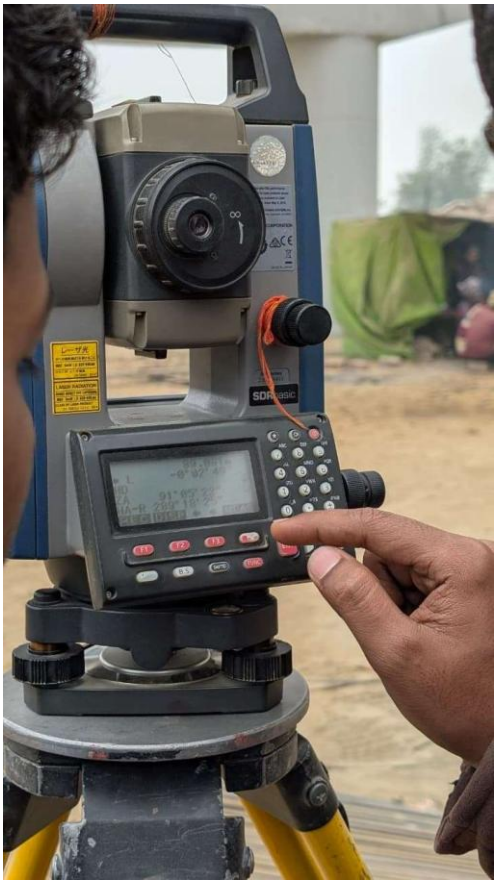
Layer Composition (Top to Bottom):

Layer Name	Material	Approx. Thickness	Function
Surface Layer	**PQC** (Pavement Quality Concrete)	**280 mm - 300 mm**	High-strength cement concrete (M40/M50 grade); provides rigid support.
Separation Layer	**Polythene Sheet** (125 micron)	**N/A**	Reduces friction between PQC and DLC; prevents moisture movement.
Base Course	**DLC** (Dry Lean Concrete)	**150 mm**	Low-strength concrete (M10/M15); acts as a stable, non-erodible base.
Sub-Base	**GSB** (Granular Sub-Base)	**200 mm - 250 mm**	Drainage layer to prevent pumping of fines under the slab.
Subgrade	**Compacted Soil**	**500 mm**	Foundation (CBR ≥ 8%).

Cross Section Features:
Joints: Contraction joints (with dowel bars) and longitudinal joints (with tie bars) are essential.
Texture: Texturing (brushing/tining) is done on wet PQC for skid resistance.



Components of Total Station



Total Station

Contractual Agreement

Overview of Contractual Agreement

1. Project Framework The project is executed under the **Engineering, Procurement, and Construction (EPC)** mode, which is the preferred contract model for National Highways under the **Bharatmala Pariyojana**. The agreement was signed between the **National Highways Authority of India (NHAI)** (referred to as the "Authority") and **Vijai Construction (India) Private Limited** (referred to as the "Contractor").
Unlike traditional Item Rate contracts, this EPC contract mandates that the Contractor is responsible for the **Survey, Investigation, Design, and Construction** of the project, ensuring a single point of responsibility.

2. Scope of the Agreement The contract encompasses the construction of the **4-Lane Greenfield North-East side Gorakhpur Bypass**, covering a total design length of **26.616 km**.
Start Point: NH-29E (Chainage 83.308)
End Point: NH-28 (Chainage 285.100)
Location: Gorakhpur District, Uttar Pradesh.

3. Key Contractual Data

Parameter	Details
Contract Mode	EPC (Engineering, Procurement, & Construction)
Authority	National Highways Authority of India (NHAI)
Contractor	Vijai Construction (India) Pvt. Ltd.
Project Type	Greenfield Alignment (New Construction)
Lane Configuration	4-Lane Divided Carriageway with Paved Shoulders
Pavement Design	Flexible (Bituminous) & Rigid (Concrete at Tolls/Structures)

4. Obligations of the Contractor Under **Article 3** of the agreement, the Contractor's obligations are comprehensive and include:
Engineering Design: Preparation of the Good for Construction (GFC) drawings, Plan & Profile, and structural designs based on site investigations.
Procurement: Sourcing all materials (cement, bitumen, steel, aggregates) and machinery required for the works.
Construction: Execution of the works in conformity with **MoRTH Specifications** and **IRC Codes**.
Safety & Traffic Management: Ensuring safety zones and traffic diversion during construction.

5. Maintenance and Defects Liability Period (DLP) A critical aspect of this agreement is the long-term quality assurance provided by the Contractor. As per **Clause 17.1**, the Contractor is liable for rectifying defects for a specific period after the provisional completion certificate (PCC) is issued:
Flexible Pavement (Roads): 5 Years.
Rigid Pavement (Toll Plazas) & Major Bridges: 10 Years.
Culverts & Other Structures: 5 Years.
During this period, any structural or functional failure (such as potholes, rutting, or cracking) must be repaired by the Contractor at their own cost.

6. Payment Mechanism Since this is an EPC contract, payments are not based on individual quantities (like per ton of bitumen) but are linked to **Payment Milestones**. The total contract price is divided into percentages assigned to specific stages of work (e.g., 5% on completion of Earthwork, 10% on completion of GSB, etc.).

Legal & Statutory Framework

Legal Framework and Land Acquisition

1. Overview The execution of the 4-Lane Greenfield Gorakhpur Bypass is governed by the statutory framework of the Government of India. The primary legislations enabling the land acquisition, authority ownership, and compensation for this project are the **National Highways Act, 1956** and the **RFCTLARR Act, 2013**.

2. The National Highways Act, 1956 This is the principal act that empowers the Central Government to declare, develop, and maintain highways. For the Gorakhpur Bypass project, this Act provides the legal authority for the **National Highways Authority of India (NHAI)** to operate.
Declaration of Highways (Section 2): Empowers the government to declare specific roads as National Highways.
Land Acquisition (Sections 3A to 3J): This is the most critical section for the project's Right of Way (ROW).
Section 3A: Notification of intention to acquire land.
Section 3D: Final declaration of acquisition (vesting land with the Central Govt).
Section 3G & 3H: Determination and deposit of compensation amounts for landowners.
Execution Models (Section 8A): This section allows the Central Government (or NHAI) to enter into agreements with private parties for development. This is the legal basis for the **EPC Contract** awarded to **Vijai Construction (India) Pvt. Ltd.**

3. The RFCTLARR Act, 2013 *Full Name: The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013.*
While the NH Act outlines the *process* of acquisition, the RFCTLARR Act, 2013 ensures that the *compensation* and *treatment* of affected families are fair. It replaced the colonial Land Acquisition Act of 1894.
Fair Compensation: The Act mandates that compensation be calculated based on the market value of land multiplied by a factor (1 or 2 depending on urban/rural location) plus a 100% solatium (extra payment for compulsory acquisition).
Rehabilitation & Resettlement (R&R): Beyond monetary compensation for land, the Act ensures support for those losing livelihoods (Schedule II), including housing units or employment allowances where applicable.
Transparency: It mandates Social Impact Assessments (SIA) to ensure the project serves a genuine public purpose with minimal displacement.
Applicability: As per **Schedule IV** of this Act, the National Highways Act, 1956 is listed as an enactment regulated by the compensation provisions of the 2013 Act to ensure uniformity.

4. Relevance to the Project For the **26.616 km Gorakhpur Bypass**:
The **ROW (Right of Way)** handed over to the Contractor was acquired using the powers of **Section 3D of the NH Act, 1956**.
The compensation paid to the farmers and landowners in Gorakhpur District was determined using the valuation principles of the **RFCTLARR Act, 2013**.
The "Encumbrance-Free Site" required for the commencement of civil works (Earthwork/Pavement) is a direct result of these successful legal procedures.

SAFETY MEASURES

SAFETY MEASURES

1. Introduction

Highway construction safety measures encompass a wide range of precautions, including
Personal Protective Equipment (PPE), proper traffic control, and equipment maintenance. Workers must wear appropriate PPE, including hard hats, safety glasses, and reflective
clothing, and avoid blind spots when operating machinery. Clear signage and markings, including cones, barriers, and reflective tape, are crucial for delineating work zones and
guiding traffic. Regular equipment inspections and preventative maintenance are essential to prevent malfunctions and accidents, as is adhering to traffic laws and speed limits.

2. Importance of Safety Measures

1. **Protection of Workers:** Workers on highway construction sites face numerous risks, such
as falling from heights, being struck by heavy equipment, or encountering hazardous materials. Implementing safety measures such as proper training, protective gear, and
clear signage reduces the risk of injuries and fatalities.
2. **Public Safety:** Highway construction often occurs on busy roads, where the public continues to use the highway, sometimes without clear detours or alternate routes. Traffic
management, clear signs, and speed reductions are critical to ensuring that drivers, passengers, and pedestrians are not exposed to unnecessary danger from construction
activities.
3. **Compliance with Legal and Regulatory Requirements:** Many regions have strict safety regulations that must be adhered to during construction activities. These regulations are designed to protect not just the workers, but also the environment and surrounding communities. Failure to comply can lead to fines, legal actions, or project delays.

3. **Reduction of Delays and Costs:** Accidents on construction sites can cause delays, increase insurance costs, and damage the reputation of construction firms. By ensuring safety, contractors can avoid these disruptions and ensure the project stays on track and within budget.
4. **Long-Term Structural Integrity:** Safety measures often extend to the construction of the highway itself. Using proper construction techniques and quality control ensures that the road is durable and safe for public use, reducing the need for costly repairs or maintenance in the future.
6. **Positive Public Image:** Construction firms that prioritize safety are viewed more favourably by the public. Demonstrating a commitment to safety builds trust with local communities, stakeholders, and clients, potentially leading to more business opportunities in the future.
7. **Environmental Protection:** Certain safety measures also have an environmental aspect. For example, ensuring that the construction process does not cause undue pollution or harm to local ecosystems is a key part of maintaining safety standards.

3. Various Safety Measures Taken

1. Implement Proper Training

As with any new environment, those working on road or highway construction sites should
go through proper training to be aware of the policies in place and the imminent hazards they'll face. Relaying consistent protocols to your team also ensures that everyone follows the same procedures. Your training should address things like:

1. Safety training schedules
2. Traffic control plans
3. Schedules for equipment and materials inspections
4. Known hazards for your worksite and plans to alleviate them
5. Equipment training
6. Emergency and first aid plans

In addition to initial training, you should also brief your crew every day since conditions and hazards can greatly vary.

6.3.2 Ensure a Competent Person is On Site

A competent person as someone who can spot existing and potential hazards in and around the work area that are dangerous to employees. Competent people are also authorized to make changes to the worksite to remove identified dangers.

Requiring a competent person at all times seems simple, but can quickly be derailed if your only competent person needs to temporarily leave the worksite. Identify your key "competent people" in your crew and implement a schedule, if needed, to ensure that at least one person is on site at all times.

6.3.3 Set Up a Proper Perimeter

The worksite should have ample space on all sides for work to be done. Using proper barriers, cones and other barricades ensures drivers know when the worksite begins and ends. Barricades should also be used inside the worksite to indicate where it's safe for crew members to walk, where materials are stored and where equipment is used. Your crew should also mark utility lines to prevent electrocutions and other avoidable injuries.

Figure 6.1 Usage of Barrier

6.3.4 Increase Worker and Work Site Visibility

OSHA requires construction employees working on highway and road sites to wear high-visibility clothing. High-visibility clothing includes things like hats, vests and armbands with reflective or fluorescent materials. You can also use lights to increase visibility if your crew is working in the evening or during periods when it's difficult to see.

6.3.5 Wear the Proper Safety Equipment

Figure 6.2 Wearing Proper Safety Equipment

Additionally, crew members should also wear earmuffs or earplugs when working in areas with extreme noise. Extreme construction noise can cause severe hearing damage if precautions aren't taken by your crew. Other important pieces of personal safety equipment include steel-toed shoes, gloves and respirators. Crew members should also ensure everything fits properly and do checks throughout the day to confirm everything is in place.

6.3.6 Control Traffic

Post traffic signs an ample amount away from the work zone so drivers can prepare to adjust to changes like slowdowns or reduced lanes. You should also give drivers enough space to safely transition out of the work zone and back into the regular flow of traffic. Flaggers who are directing vehicles should never turn their backs to oncoming traffic.

6.3.7 Use Caution When Operating Heavy Machinery

Large machines, like rollers and excavators, have blind spots and are capable of fatally injuring your fellow crew members. Fortunately, there are a number of things you can do to safely operate heavy equipment on a road or highway construction site.

Safety while operating Machines

a) Use a seatbelt at all times
b) Apply parking brakes whenever vehicles are not in use
c) Put a block in front or behind of the tires when left on an incline or decline
d) Use a spotter when moving, unloading and loading equipment
e) Check that mirrors and other visual aids like tail lights are attached and operational

CONCLUSION

The internship training carried out under the Project Implementation Unit (PIU), Gorakhpur, of the National Highways Authority of India (NHAI) provided valuable practical exposure to the planning, execution, quality control, and management aspects of highway infrastructure projects. The training successfully bridged the gap between theoretical knowledge gained in the classroom and real-time field practices followed in large-scale national highway projects executed under the EPC mode.

Throughout the training period, exposure was gained to various quality assurance and quality control (QA/QC) laboratory tests related to soil, aggregates, bitumen, and concrete, which are critical for ensuring durability and performance of pavements and structures. Tests such as CBR, Atterberg limits, Free Swell Index, Aggregate Impact Value, water absorption, field dry density, slump test, and strength tests of concrete helped in understanding the acceptance criteria, importance of specifications, and consequences of deviation from prescribed limits.

Field exposure to pavement construction activities, including the laying of GSB, Dry Lean Concrete (DLC), and Dense Bituminous Macadam (DBM), provided insight into construction sequencing, equipment usage, compaction control, and workmanship requirements. Observing these activities enhanced understanding of pavement layer interaction and the role of each layer in overall pavement performance.

The visit to the major bridge structure near Mahesra Lake offered comprehensive understanding of bridge engineering practices, including friction pile foundations, substructure components, PSC girders, post-tensioning systems, bearing arrangements, and deck slab construction. Exposure to structural drawings and prestressing operations helped in correlating design concepts with actual site execution.

Work related to the alignment and erection of Reinforced Earth (RE) walls provided practical knowledge of surveying techniques using a total station, resection method, levelling pad construction, RE panel erection, geosynthetic reinforcement, drainage provisions, and stability considerations. This experience highlighted the importance of precision in surveying and systematic construction methodology for retaining structures.

Additionally, an overview of the EPC contractual agreement improved understanding of how technical execution on site is governed by contractual obligations, quality requirements, timelines, and responsibilities of various stakeholders involved in highway projects.

Overall, the internship proved to be a highly enriching learning experience that strengthened technical knowledge, enhanced practical skills, and developed professional understanding of highway engineering projects under NHAI. The training has provided a strong foundation for future professional practice in the field of civil and highway engineering.